

DevOps Series

Using Ansible *to* Build GNU Guile

This is the 22nd article in the DevOps series and here we will use Ansible to automate the process of building GNU Guile from its source code. GNU ‘Guile’ is an acronym for GNU Ubiquitous Intelligent Language for Extensions, which is an implementation of the Scheme programming language and an extension language for the GNU Project.

GNU Guile was designed by Aubrey Jaffer, Tom Lord and Miles Bader and was first made available in 1993. You can write desktop, Web and command line applications in GNU Guile. The latest stable version, as of July 2018, is v 2.2.4. It is released under the GPL and is available for both IA-32 and x86-64 platforms.

Setup

A Parabola GNU/Linux-libre (x86_64) 2018.06.02 ISO is used to set up a guest virtual machine (VM) using KVM/QEMU. The host system is also a Parabola GNU/Linux-libre x86_64 system and Ansible is installed using the distribution package manager. The version of Ansible used is 2.6.0 as indicated below:

```
$ ansible --version
ansible 2.6.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.6/site-
packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

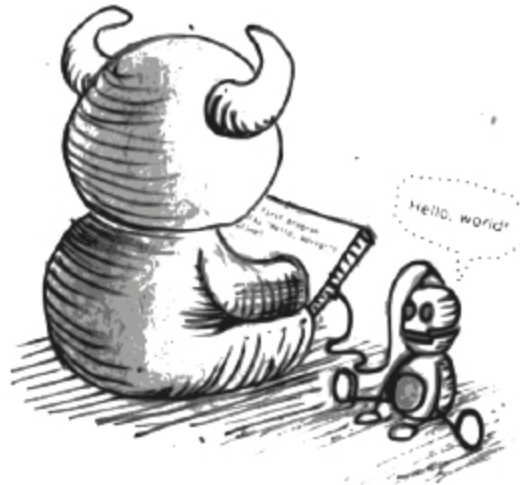
The Ansible inventory and playbook files are created on the host system, as follows:

```
ansible/inventory/kvm/
  /playbooks/configuration/
```

The *inventory/kvm/inventory* file contains the following:

```
guile ansible_host=192.168.122.109 ansible_connection=ssh
ansible_user=parabola ansible_password=password
```

The ‘sudo’ package needs to be installed in the guest Parabola VM. A Parabola user is created in the guest VM and sudo access is provided for this user using the ‘visudo’ command. Python and Openssh software are also required to be installed on the guest VM. The ssh daemon is started in the



guest VM using the following command:

```
$ sudo systemctl start sshd.service
```

You should add an entry in the */etc/hosts* file on the host system for the Parabola guest VM, as indicated below:

```
192.168.122.109 guile
```

You can now test the connectivity from Ansible to the Parabola guest VM using the following Ansible command:

```
$ ansible -i inventory/kvm/inventory guile -m ping
guile | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Software dependencies

GNU Guile requires a number of software dependencies, such as *gmp*, *libunistring*, *libffi* and *readline*. The essential build tools—*autoconf*, *automake*, *libtool*, *flex*, *gcc*, *gc* and *make* — are also needed for building GNU Guile. The Ansible playbook to install the above software is given below:

```
---
- name: Repository setup
  hosts: guile
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [packages]

tasks:
  - name: Install dependencies
    package:
      name: "{{ item }}"
      state: latest
    with_items:
      - "git"
      - "gmp"
      - "libunistring"
```